Semantic interference in LLM attention: Evidence from relative clauses

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Introduction. Recent research suggests that the attention mechanism of Transformer-based large language models (LLMs) operates as a kind of cue-based retrieval processor that parallels human working memory. Specifically, attention patterns that guide next-word prediction have been shown to index similarity-based interference in human sentence processing, linking memory- and expectation-based processes within a single architecture [1,2]. While these findings are compelling, prior work has focused narrowly on syntactic effects, such as the classic processing asymmetry between subject and object relative clauses (SRC/ORC), difficulty with center-embedding, and agreement attraction. It is an open question whether attention can capture the more nuanced semantic properties that shape memory interference in human sentence processing. The current study addresses this question by testing whether attention is sensitive to the referential class distinctions that modulate similarity-based interference in relative clause processing [e.g., 3,4]. Results demonstrate that attention metrics fully capture classic behavioral profiles, providing new evidence that Transformers function, in part, as a cue-based retrieval sentence processor that is sensitive to both syntactic and semantic properties in memory.

Behavioral effects of interest. A classic finding in human sentence processing is that the advantage for SRCs over ORCs is reduced or eliminated when nouns 1 and 2 differ in referential class (e.g., definite description vs. proper name, pronoun, quantified pronoun) [3,4]. Such effects have been attributed to a reduction in similarity-based interference [3-5]. Recently, [2] demonstrated that attention metrics derived from GPT-2 capture the basic structural SRC/ORC contrast. The present study extends this work by testing whether attention is also sensitive to the subtle semantic features of referential class that modulate relative clause processing in humans.

LLM modeling. To assess this, three attention-based metrics were analyzed in a Bayesian framework at the same critical regions defined in the behavioral literature (i.e., ORC verb and SRC object noun): (i) attention shift, (ii) attention-to-target, and (iii) attention entropy. All stimuli from [3,4] were tested, which manipulated structure (SRC vs. ORC) and referential class (definite description, indefinite, generic, superordinate RC subject, superordinate main subject, pronoun, name, quantified pronoun) (Table 1). Attention-based metrics captured the classic SRC/ORC asymmetry with robust sensitivity to referential class: Attention shift was larger in ORCs than SRCs for the baseline definite condition as well as the indefinite, generic and superordinate conditions ($P(\beta > 0) > .96$ in all cases). Crucially, this difference was reduced or eliminated when referential class differed (pronoun, name, quantified pronoun conditions) ($P(\beta > 0) = .60$-$.80$); all interactions: $P(\beta > 0) > .94$) (Fig. 1). Attention-to-target also increased when referential class differed relative to the baseline definite condition ($P(\beta > 0) > .95$ in all cases) (Fig. 2), indicating reduced interference. Attention entropy was also lower (more focused retrieval) when referential class differed ($P(\beta > 0) = 1$ in all cases) (Fig. 3), again indicating reduced interference

Discussion. These results fully recapitulate the behavioral profiles reported in [3,4] demonstrating that attention metrics are sensitive to not only structural contrasts but also subtle semantic properties. These findings bolster the recent proposal that attention can be used to test predictions of the cue-based retrieval theory [2], which seeks to provide a uniform treatment of both syntactic and semantic interference in human sentence processing. More broadly, they provide new computational evidence that the semantic features that induce interference can be learned from self-supervised task of next-word prediction [1,2]. Together with [2], we can address a key challenge for the cue-based retrieval theory of human sentence processing: current models like ACT-R rely on postulated binary cues (e.g., ±shatterable; [6]). However, the differences in referential class that drive the semantic effects in relative clause processing cannot be defined in binary terms. Since LLMs learn the relevant (non-binary) features from distributional patterns for next-word prediction, they provide a more flexible framework for modeling how abstract semantic properties guide encoding, retrieval and prediction, contributing to recent efforts to unify memory- and expectation-based processes within a single framework [e.g., 2,7,8].

Table 1: Sample experimental stimuli from [3,4]. Critical regions in bold.

Definite description (baseline condition)	ORC	The banker that the barber **praised** climbed the mountain …
	SRC	The banker that praised the **barber** climbed the mountain …

Indefinite: *a barber* **Generic:** *barbers* **Superordinate noun:** *the person*
Pronoun: *you* **Name**: *Sue* **Quantified pronoun:** *everyone*

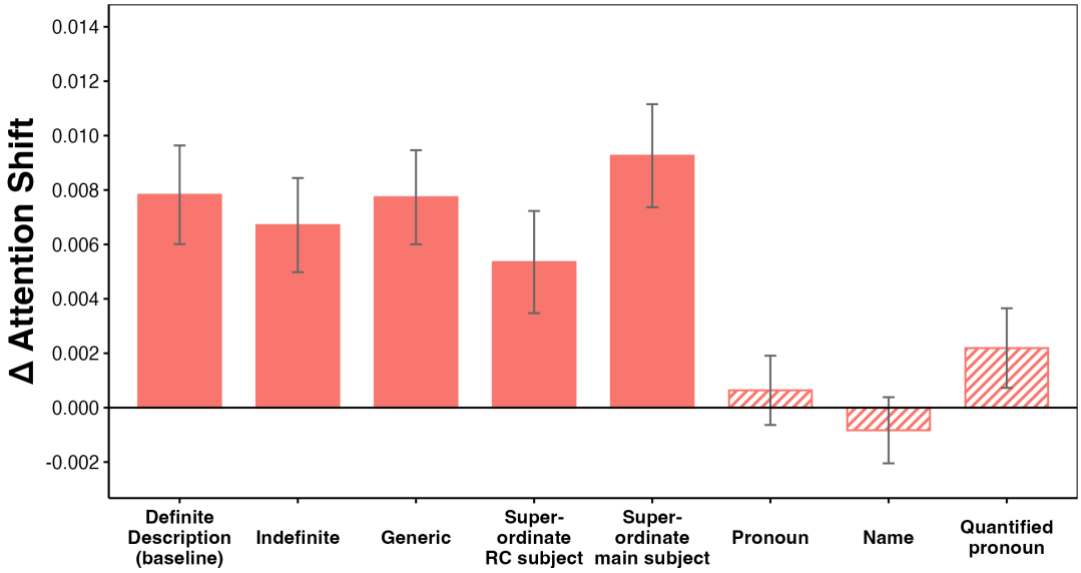
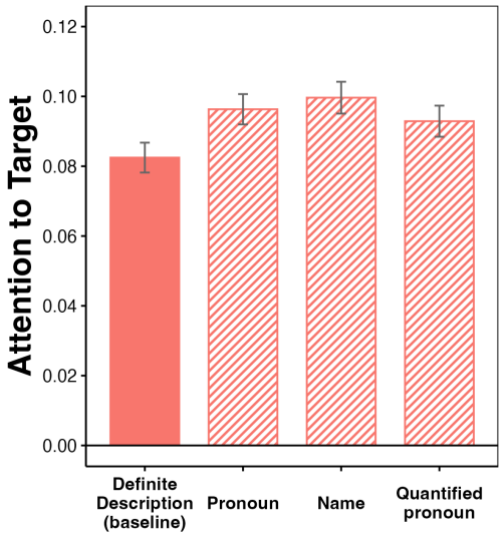


Fig. 1. Positive values indicate greater attention weight for ORCs compared to SRCs, reflecting difficulty. Solid bars are predicted to show the ORC/SRC asymmetry and striped bars are predicted to show a reduced effect as shown in [3,4].



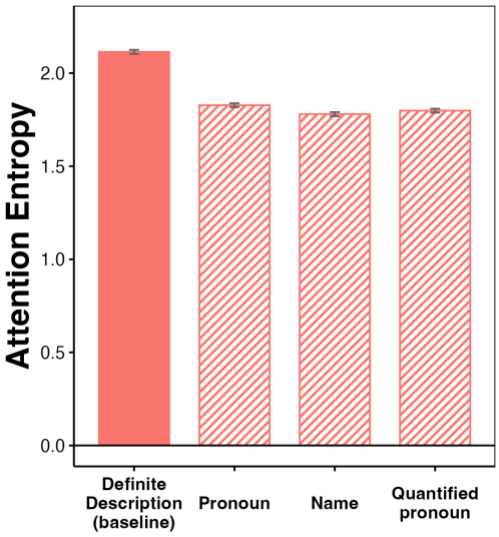


Fig. 2 (left). Higher values indicate greater attention allocated to the target noun, reflecting more accurate and focused retrieval. **Fig 3 (right).** Higher values reflect more diffuse attention over the prior context, indicating greater uncertainty about the target during retrieval.

References. [1] Ryu & Lewis. 2021. CMCL. [2] Ryu & Lewis. 2025. JML. [3] Gordon et al. 2001. JEP. [4] Gordon et al. 2004. JML. [5] Van Dyke & McElree. 2006. JML. [6] Cunnings & Sturt. 2018. JML. [7] Futrell et al. 2020. CS. [8] Hahn et al. 2022. PNAS.